F6: local search and annealing on the m = 2 decomposition landscape

Run F6-RUN-01 of study `semaev_qubo_annealing_20261110`. Class: boundary diagnostic; no S, no rho row, no leaderboard change. All measured numbers below are `[pcc:unverified]` (no pcc claim id exists for this run yet).

Question

Does any local search or annealing method find a planted decomposition $R = P_1 + P_2$ with $x(P_i) \le B$ in fewer cost evaluations than the exact baselines (box enumeration $\Theta(B^2)$, root-solving enumeration $\Theta(B)$), with an exponent in B below theirs? (Moonshot G1.)

Mechanism

State $(x_1, x_2) \in [1, B]^2$; cost $c = \min(v, p - v)$, $v = S_3(x_1, x_2, x_R) \bmod p$; $c = 0$ iff a decomposition exists at those abscissae (checked by lifting and adding). Moves: bit flips of a Gray-coded coordinate, ±1 steps, random restart. If $S_3 \bmod p$ is pseudo-random off the solution set there is no usable gradient and local search reduces to sampling.

Method

p = largest prime ≡ 3 (mod 4) below 2^n, $n \in \{12, 16, 20, 24, 28\}$, one prime-order curve per n; $B = \lceil \sqrt{p} \rceil$; 20 planted instances per n (seed `F6/inst/<n>/<i>`). Solvers: uniform random sampling (control), greedy with restarts, simulated annealing at three schedules (sa_B, sa_4B, sa_16B), parallel tempering (8 replicas); baseline root_enum counts x_1 values tried. Metric: cost evaluations to the first verified $c = 0$, cap $\min(50 \cdot B^2, 2 \cdot 10^9)$ (§6 amendment). Exponent: ln-ln OLS of the median on B, bootstrap 95% CI over instances (1000 reps), over all n and over $n \ge 20$. Landscape: lag-1..8 autocorrelation along bit-flip walks vs a random landscape with the same marginal, $n \le 20$. Native Rust tool, local host (4 vCPU), counts only.

Results `[pcc:unverified]`

Median evaluations over 20 instances (censored count in brackets); p, B, B^2, mean box solutions per n.

n	p	B	B^2	box sol.	uniform	greedy	sa_B	sa_4B	sa_16B	pt	root_enum
12	4091	64	4,096	2.80	922 [0]	2,466 [0]	3,636 [0]	2,354 [0]	5,162 [0]	4,783 [0]	25 [0]
16	65519	256	65,536	3.40	13,792 [0]	35,242 [0]	59,672 [0]	89,760 [0]	71,861 [0]	33,464 [0]	42 [0]
20	1048571	1,024	1,048,576	2.75	191,126 [0]	394,924 [0]	2,466,308 [0]	1,722,998 [0]	893,260 [0]	939,996 [0]	290 [0]
24	16777199	4,096	16,777,216	3.20	3,898,006 [0]	4,346,316 [0]	19,692,566 [0]	13,430,037 [0]	34,937,164 [0]	30,191,004 [0]	640 [0]
28	268435399	16,384	268,435,456	3.30	40,138,032 [0]	66,891,984 [0]	820,856,178 [2]	765,670,044 [2]	367,805,855 [0]	395,638,322 [2]	3,324 [0]

solver	e [95% CI], all n	e [95% CI], n ≥ 20
uniform	1.948 [1.790, 2.115]	1.929 [1.572, 2.251]
greedy	1.820 [1.668, 2.003]	1.851 [1.614, 2.105]
sa_B	2.197 [2.050, 2.321]	2.095 [1.761, 2.440]
sa_4B	2.192 [1.993, 2.372]	2.199 [1.875, 2.486]
sa_16B	2.058 [1.835, 2.304]	2.171 [1.593, 2.474]
pt	2.124 [1.981, 2.325]	2.179 [1.791, 2.506]
root_enum	0.901 [0.797, 1.038]	0.879 [0.626, 1.135]

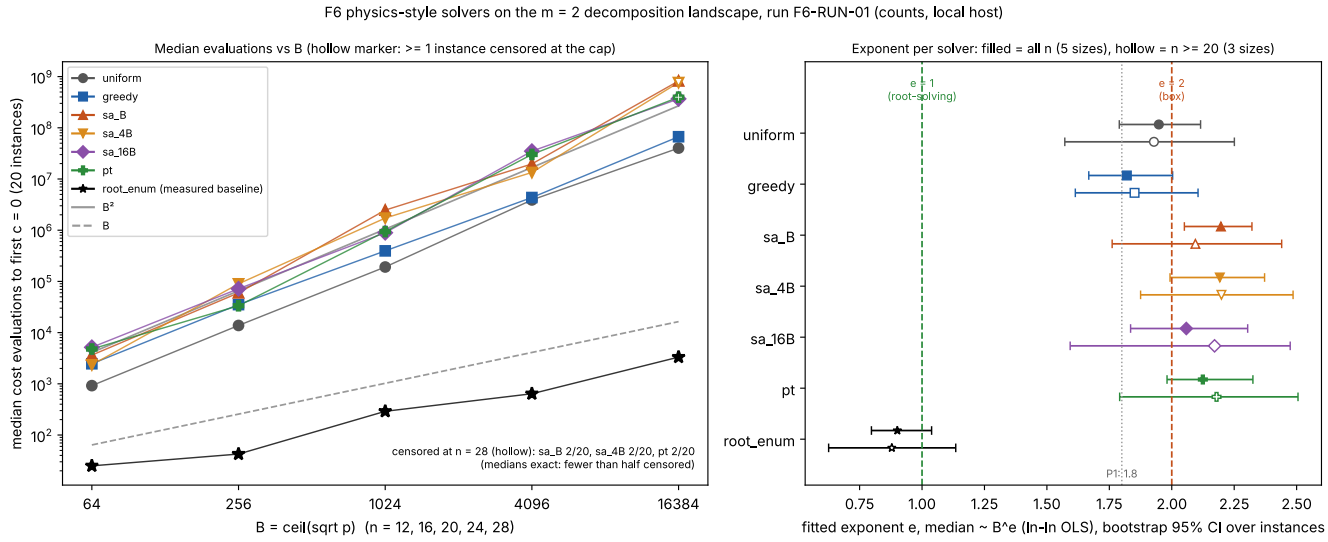


Figure 1: Left: median cost evaluations to first c = 0 vs B (log–log) per solver, with B^2 and B reference lines and the measured root_enum; hollow markers: at least one of 20 instances censored (n = 28: sa_B, sa_4B, pt, 2/20 each). Right: fitted exponent e with bootstrap 95% CI per solver, all n (filled) and $n \ge 20$ (hollow), with e = 1 (root-solving), e = 2 (box) and the P1 threshold 1.8. Run F6-RUN-01, 20 instances per n, local host. Source: `results/run.json`; `figures/make_figure.py`.

Exponents: uniform 1.948 (n ≥ 20: 1.929); greedy 1.820 (1.851); sa_B 2.197 (2.095); sa_4B 2.192 (2.199); sa_16B 2.058 (2.171); pt 2.124 (2.179); root_enum 0.901 (0.879). Autocorrelation at lag 1 is in band at n = 12, 16, 20 (walk means −0.0006, +0.0003, −0.0012; bands

about ±0.094); the lag-2 mean excess over the random landscape is recorded (advisory z 49.6, 36.6, 26.3) and not scored. Costs: 0 verification failures, 0 instrument errors, 0 spurious zero-cost states. [pcc:unverified]

Reading (PROTOCOL.md §4)

1. Instrument failure: not triggered. P3: uniform e = 1.948, within 2 ± 0.25; 0 verification failures; 0 instrument errors. [pcc:unverified]
2. Lead alive: not met. Smallest n ≥ 20 upper CI over the six solvers is 1.614 (greedy), none below 1.0; P2 in band at lag 1 at all three sizes (0 failing sizes, rule needs ≥ 3). [pcc:unverified]
3. Rule 3 applies, verbatim: "Closed: otherwise: no local-search method beats root-solving enumeration on the `m = 2` decomposition landscape at these sizes." [pcc:unverified]

P1 (e ≥ 1.8 for every solver): held under the protocol's wording, by point estimates; greedy is the margin (1.820 all n, 1.851 n ≥ 20); its CIs [1.668, 2.003] and [1.614, 2.105] contain 1.8 and the protocol froze no CI test. P2: held under the in-band convention at all 8 lags, with the lag-2 and lag-4 advisory z noted above. P3: held ($|1.948 - 2| = 0.052 \le 0.25$). No solver reached e ≤ 1 (prior 0.005). [pcc:unverified]

Cap note

The §6 amendment sets the cap to $\min(50 \cdot B^2, 2 \cdot 10^9)$; the tool applies it (every censored row records 2,000,000,000). The stdout header ("cap 50 B^2"), `header.eval_cap` and `summary.baselines_per_n[].cap_50B2` (13,421,772,800 at n = 28) still carry the unamended label, so at n = 28 the labelled cap differs from the applied one; for n ≤ 24 they coincide. Six of 140 n = 28 rows are censored: sa_B, sa_4B, pt, two instances each; the medians there are exact (fewer than half censored). [pcc:unverified]

Scope

m = 2 only, these solvers only, n ≤ 28 (landscape n ≤ 20); no QUBO hardware. The §9a isolab-eks confirmatory run is open; the host was shared; no timing is a result. The tool ran with 4 threads (no `--threads` flag); results are seeded per instance and solver, so by the tool's design they do not depend on the thread count (an assertion of design, not measured). The research-kb server was unavailable; no novelty claim is made.

Provenance

Card	`CARD-semaev-qubo-annealing-20261110-F6-RUN-01`, sha256 `2ba610affa718256691e81900195fff1b7c6a9859b44ef8c6a8e7a2d72bd9ece`
Commit	`8c3b356ff45ec268ea7d0a21f28050200aa4c515` (8c3b356ff4)
Receipt	sha256 `7fce30d49275716076fa84a840441a24e896bd4d95059491388714e481fd1920`; status completed, 13 commands; tree dirty before and after (other studies, card outputs)
Binary	sha256 `07371fffa1bb890de18d3d000ee5418987499e01bc78e9526f96a6653a3d9cae` (receipt); the card's pre-build `binds` entry lists `c70c8310f6752cd64f326c6feb8aa625f5b9a556e6b0ea93a535a250c187ba9f`, not reconciled
run.json	sha256 `9bbea2ea5dd017b85461ce9fc619cf99768bcdc2cdfb0a0a9e58586d786b3cdd`
FREEZE.sha256	PROTOCOL.md `7c427397febd48f48ba9941152997628ab6f0ec308fcb0dc7d550c12222387eb`; tool/src/main.rs `88d7dd340c8b48164696db4d0a670537e1e574c3d32938f3d0c8fbd540dad61f`; tool/Cargo.toml `69fa9a7ee61e4e8814462bb57765f56295802a455d1795a631936c702fc8f61f`; tool/Cargo.lock `a1e70b445783b45e612a8ab1c862018b98769babefefb8c293684f7ec1c94aea`
Report build	typst Python wheel 0.15.2 (scratch venv); figure by matplotlib 3.11.2